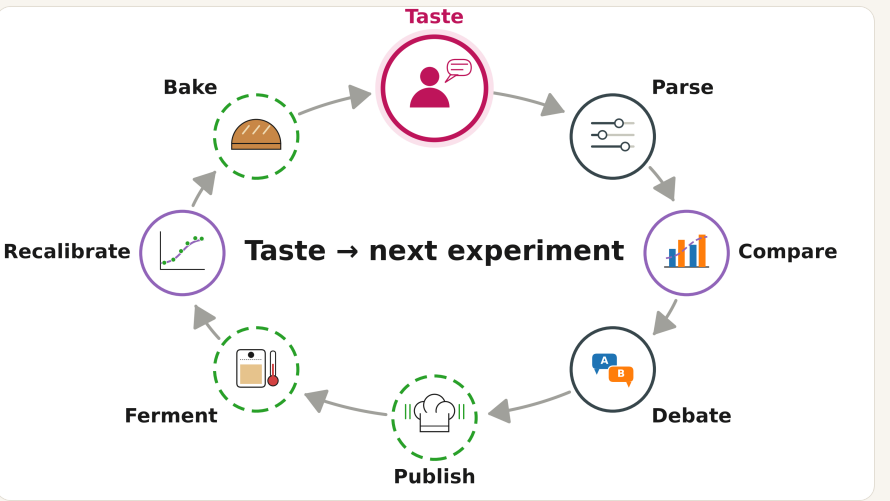
FERMENTAI · SCIENCECLAW 2026

# Every kitchen can become a food-science lab

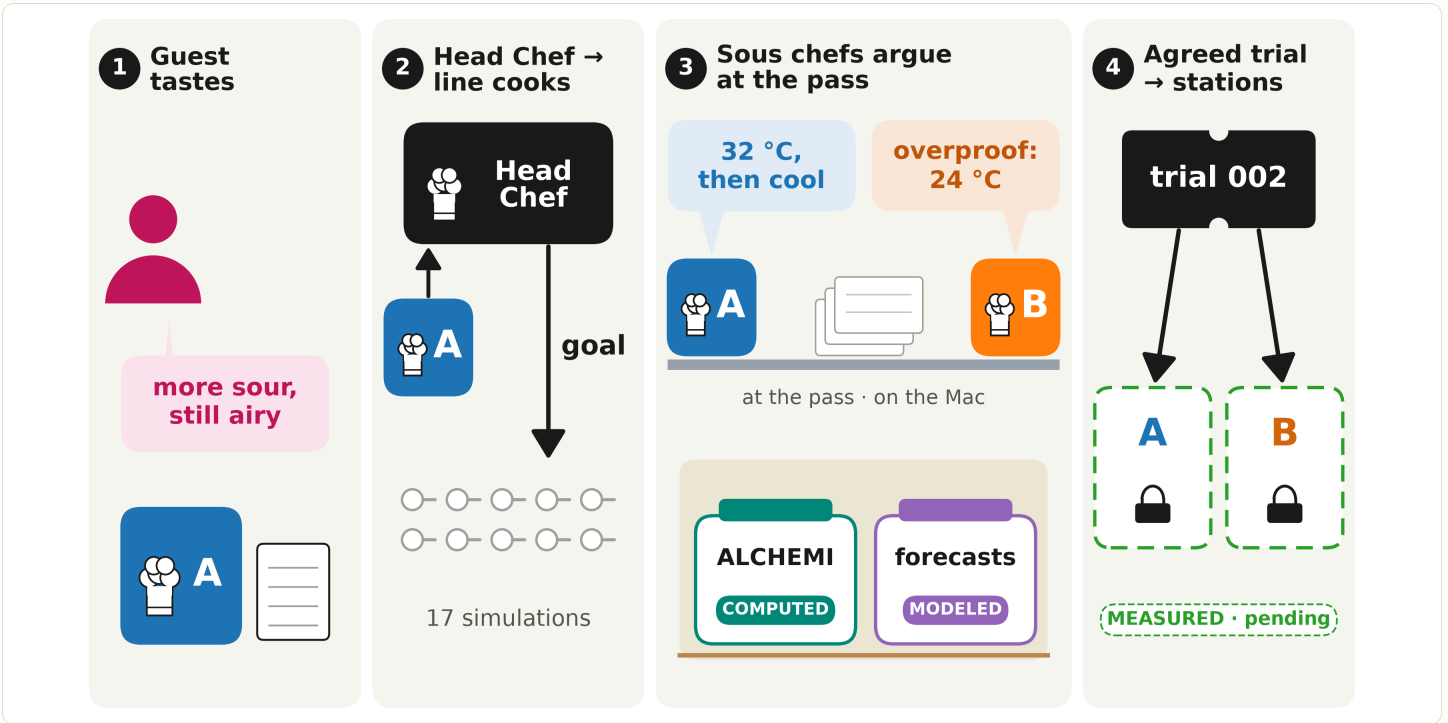
- **Problem:** flavour by guesswork
- **Vision:** every kitchen a lab
- **Why now:** cheap GPU chemistry + AI agents




## SLIDE 1 / 4 • COVER • SPEAKER NOTES

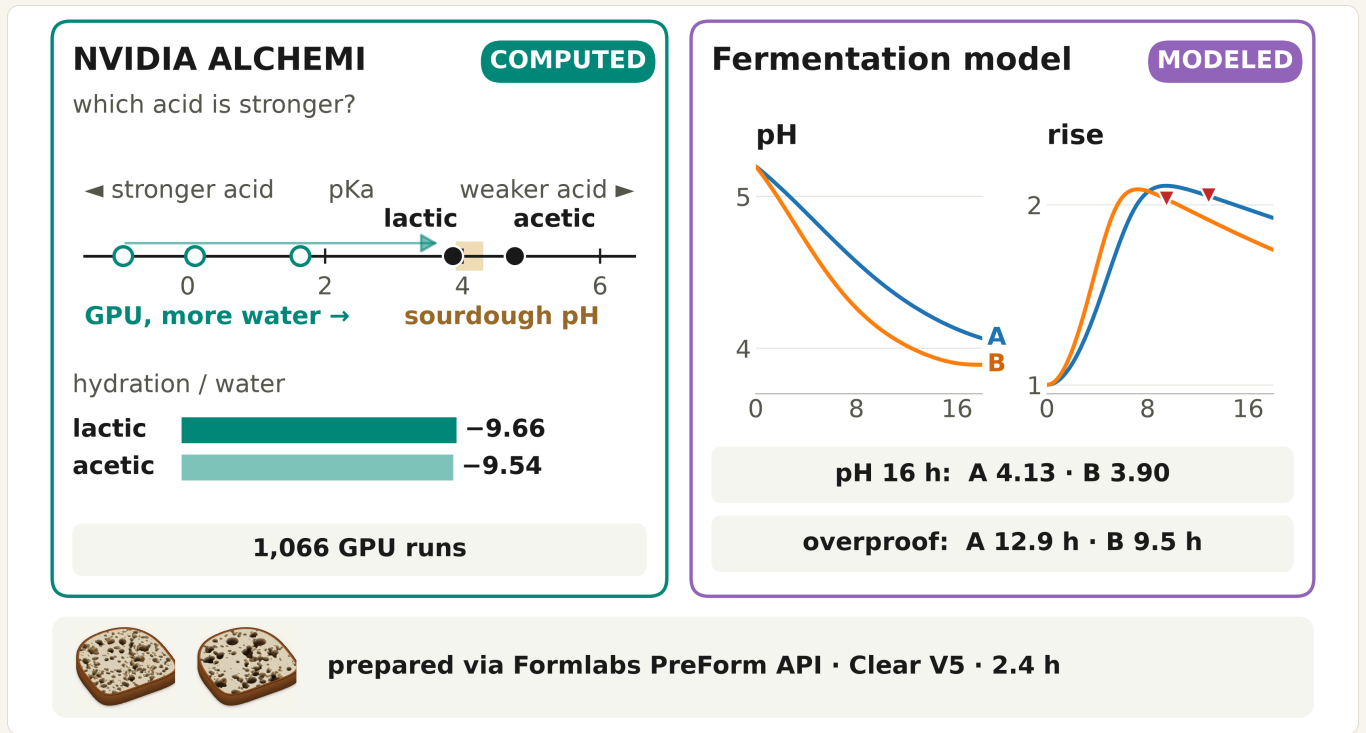
Every sourdough baker runs an experiment every day, then throws the data away. Flavour depends on temperature over a twelve-to-sixteen-hour ferment, and people tune it by feel. Our vision is that every kitchen becomes a lab, with AI scientists that run experiments, share evidence and learn from the people who eat the result. Why now? Molecular chemistry on a GPU costs cents, sensors cost dollars, and AI agents can reason and argue. This loop is the product. You bake, you taste, you tell the system what you think, and it designs the next experiment.

# A head chef, two sous chefs, two stations




Think of it as a kitchen brigade. The guest tastes and tells Sous Chef A what they think. The Head Chef sets the dish, more sour but still airy, and sends out seventeen line cooks: candidate simulations. The two sous chefs (Scientist A and B), each running their own station, argue over the results across the pass. They draw on the same pantry: computed acid chemistry from NVIDIA ALCHEMI and our modeled fermentation forecasts. When they agree on a trial, each station gets its schedule. Every station is safety-locked, because firmware clamps each setpoint.

# Two partners: NVIDIA chemistry, Formlabs fabrication

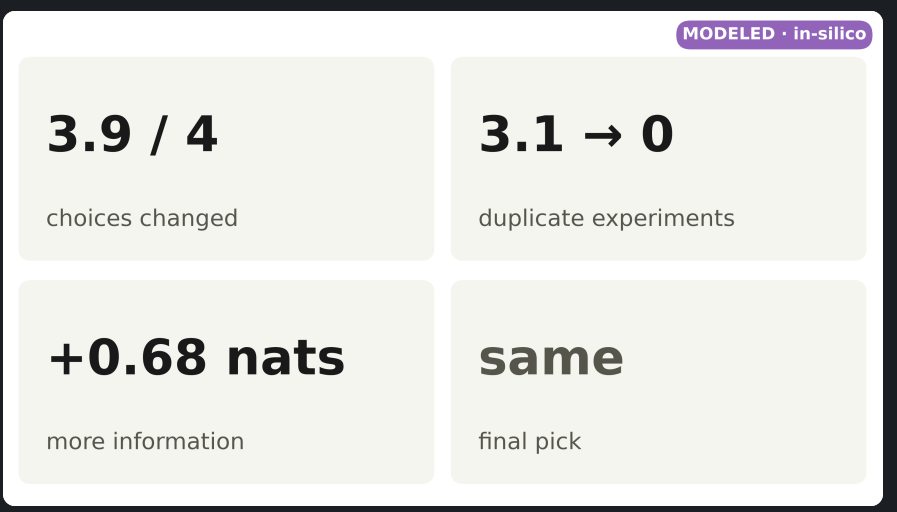





NVIDIA ALCHEMI gives us computed molecular evidence: over a thousand GPU runs, cached so the demo runs offline. It finds lactic the stronger acid, by four pKa units against one in experiment, and it moves toward experiment as we add water. That only bounds the model's acid terms. The model is built from published sourdough kinetics, Gänzle 1998 and Rosso 1993, with every parameter cited or labelled an assumption. It takes the recipe, the temperature schedule and the starter's state, and predicts pH, acids, gas, rise and overproofing. Formlabs makes the prediction physical: the PreForm API prepares both crumb specimens for print, and a person presses print.

## Working today, and what we need next

- **Computed:** GPU chemistry · print prep
- **Modeled:** forecasts · crumb · benchmark
- **Ask:** 3 pilot kitchens + GPU & printer credits




**SLIDE 4 / 4 • TRACTION • SPEAKER NOTES**

Working today: the software, two agents and a fully offline demo. The NVIDIA and PreForm results are computed. The forecasts and crumb shapes are modeled, not yet validated against a bake, and nothing is measured yet, because the chamber firmware is compiled but unproven on hardware. In a modeled benchmark, agents that talk change their choice almost every round, with a modest information gain and the same final pick. Next come measured bakes, photo validation and a network of kitchens. We’re looking for three pilot kitchens or bakeries, plus GPU and printer credits, to run the first multi-kitchen experiment. Please taste, and tell the Sous Chef at the laptop what you think.